5.5.3 Deleting with a range query

We may wish to delete records that fall in a certain range: movies with ratings between 3.5 and 4.5, flights canceled between two dates, and so on. We can use a `range` query to set a criterion for a range of values for such requirements. The following listing uses `_delete_by_query` to delete movies that earned between $350 million and $400 million.

Listing 5.37 Deleting all movies with a range of gross earnings

```
POST movies/_delete_by_query
{
  "query": {
    "range": {
      "gross_earnings_in_millions": {
        "gt": 350,
        "lt": 400
      }
    }
  }
}
```

Here, `_delete_by_query` accepts a `range` query with a match criterion: find documents whose gross earnings fall between $350 and $400 million dollars. As we expect, all matching documents are deleted.

We can also construct complex queries. For example, listing 5.38 shows a query that constructs criteria to delete movies directed by Steven Spielberg, where those movies are rated between 9 and 9.5 and earned less than $100 million. The listing uses a `bool` query as the request.

> **NOTE** For the following query, you need to index the movies dataset from the book's files: http://mng.bz/zXeX.

Listing 5.38 Deleting movies with complicated query criteria

```
POST movies/_delete_by_query
{
  "query": {
    "bool": {
      "must": [{
          "match": {
            "director": "Steven Spielberg"
          }
        }
      ],
      "must_not": [{
          "range": {
            "imdb_rating": {
              "gte": 9,
              "lte": 9.5
```

Matches movies directed by Spielberg

The rating should be greater than 9 but less than 9.5.